

Genomic and fragmentomic landscapes of cell-free DNA for early cancer detection

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Abstract

Genomic analyses of cell-free DNA (cfDNA) in plasma are enabling noninvasive blood-based biomarker approaches to cancer detection and disease monitoring. Current approaches for identification of circulating tumour DNA typically use targeted tumour-specific mutations or methylation analyses. An emerging approach is based on the recognition of altered genome-wide cfDNA fragmentation in patients with cancer. Recent studies have revealed a multitude of characteristics that can affect the compendium of cfDNA fragments across the genome, collectively called the ‘cfDNA fragmentome’. These changes result from genomic, epigenomic, transcriptomic and chromatin states of an individual and affect the size, position, coverage, mutation, structural and methylation characteristics of cfDNA. Identifying and monitoring these changes has the potential to improve early detection of cancer, especially using highly sensitive multi-feature machine learning approaches that would be amenable to broad use in populations at increased risk. This Review highlights the rapidly evolving field of genome-wide analyses of cfDNA characteristics, their comparison to existing cfDNA methods, and recent related innovations at the intersection of large-scale sequencing and artificial intelligence. As the breadth of clinical applications of cfDNA fragmentome methods have enormous public health implications for cancer screening and personalized approaches for clinical management of patients with cancer, we outline the challenges and opportunities ahead.

Sections

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Introduction

Despite efforts in prevention and treatment, cancer remains a leading cause of mortality worldwide¹. Cancer screening has been shown to reduce mortality for certain cancer types. The first widely used cancer screening approach, the Papanicolaou test or 'pap smear', was introduced in 1923 and involved collecting cells from the cervix using a swab and examining them under a microscope to identify abnormal cells². The pap smear is still widely used today, with current estimates suggesting that its use has led to at least an 80% reduction in both the incidence of cervical cancer and related mortality³.

Motivated by subsequent large-scale studies demonstrating benefits of cancer screening for lung cancer^{4,5}, colorectal cancer^{6,7}, cervical cancer⁸ and breast cancer^{9,10}, worldwide organizations have supported widespread screening programmes¹¹. For example, in the USA, the US Preventative Services Task Force recommends regular screening, depending on age or exposure history, for cervical cancer using human papillomavirus or Pap test, for breast cancer using mammography, for colorectal cancer with colonoscopy or stool-based tests, and for lung cancer with low-dose computed tomography (LDCT)¹². Other countries have screening programmes for individuals at increased risk for cancers of the liver, stomach, oesophagus, and head and neck (Table 1). Although most screening efforts are focused at asymptomatic populations at risk of disease, observational studies suggest that early diagnosis of cancer even in symptomatic individuals may improve outcomes¹³.

Nevertheless, many cancers are diagnosed at advanced stages, especially in regions with limited resources¹. Most cancers do not have effective screening strategies, and even for those with existing screening modalities, several challenges have emerged. In the USA, compliance has been encouraging for colorectal cancer, breast cancer and cervical cancer (60–80%), but can still be improved, and remains particularly low for lung cancer, in which <6% of individuals adhere to screening guidelines¹⁴. In addition, in these populations, false-positive results from imaging are not uncommon. For example, for lung cancer screening with LDCT, the positive predictive value is only 3.8%, indicating that only a small fraction of individuals who had a positive LDCT were ultimately diagnosed with lung cancer¹⁵. About half of all women who receive a yearly mammogram will have a false-positive result over a 10-year time span¹⁶. There is a clear unmet clinical need for improved and accessible screening tests for these and other cancers.

Improved understanding of the cancer genome^{17–19}, and the advent of massively parallel sequencing, also known as next-generation sequencing (NGS), has enabled rapid characterization of millions of cell-free DNA (cfDNA) fragments for genomic alterations at base pair resolution in healthy individuals and patients with cancer. Cancer genomes generally harbour thousands of genetic or epigenetic alterations^{20–22}, and release of cfDNA from cancer cells affords a potential biomarker of disease²³. Apoptosis, necrosis and active secretion are considered the main mechanisms responsible for generating cfDNA²⁴. cfDNA represents naturally digested DNA fragments that are small in size (generally <200 bp) and can be detected in many bodily fluids including blood, saliva, urine, stool and cerebrospinal fluid, among others²⁵. It is now generally accepted that cfDNA originates from nucleosomes, with most fragments associated with mononucleosomes (approximately ~167 bp in size) and a small fraction from dinucleosomes (approximately ~330 bp in size) or from larger fragments²⁶. Most current research for cancer detection has focused on cfDNA in the circulation using the tumour-derived component termed circulating tumour DNA (ctDNA). For healthy individuals, the source of plasma cfDNA is largely from leukocytes, but it is also derived from

megakaryocytes, erythrocyte progenitor cells, vascular endothelial cells, and hepatocytes, with small amounts from other cell types²⁷. In individuals with cancer, ctDNA derived from dying cancer cells contributes to overall cfDNA levels²⁸. ctDNA analyses were initially used for therapeutic stratification and informing clinical decisions during therapy²⁹, and now cfDNA-based tests are increasingly being explored for their potential of early detection of cancer.

In this Review, we examine the characteristics of cfDNA in healthy individuals and how these change in patients with cancer. We assess the fundamental strategies of targeted methods for cfDNA analyses that rely on the detection of somatic mutations or DNA methylation and the limitations of these approaches. We review the early proof-of-concept studies for genome-wide analysis of the cfDNA fragmentome, its origin from nucleosomes that can reflect changes in chromatin structure, and the ability to recover genomic and epigenomic information with whole-genome sequencing (WGS). Lastly, we discuss how these discoveries have facilitated the adoption of liquid biopsies and cfDNA analyses in a wide-range of clinical applications.

Characterizing cell-free DNA

Since the discovery of nucleic acids in human blood plasma³⁰, our understanding of the characteristics of cell-free DNA has evolved in parallel with technological advances facilitating the study of nucleic acids. Two primary avenues for sequencing and analysis of cfDNA have emerged, targeted and genome-wide approaches (Fig. 1). Compared to genome-wide approaches, targeted approaches require more complex laboratory assays involving enrichment of specific regions, tend to have greater technical and systematic sources of variation, are typically more expensive, and result in fewer observations used for test analyses. Novel methods are emerging for summarizing the multitude of characteristics of cfDNA fragments across the genome together with machine learning approaches for cancer detection. These include a recent explosion of genome-wide methods for analysis of cfDNA features, including for fragmentation characteristics, structural variants, somatic mutations, DNA methylation, and repetitive regions of genome (Fig. 1). Here, we review the biological basis of these recent methods and their application for early detection of cancer.

Genomic features of cfDNA

Targeted detection of somatic mutations. Targeted sequencing involves enriching the pool of DNA for fragments from specific regions of interest. The two most common methods for analyses of cfDNA have been based on sequencing of PCR amplicons, in which specific primers are designed to hybridize to and enable PCR amplification of specific target sequences, and selection of regions through hybrid capture, where oligonucleotide probes are hybridized to target sequences that are then isolated. With all targeted sequencing methods, the regions of interest must be identified a priori and, in the context of early detection of cancer from cfDNA, have largely been focused on areas containing common cancer-related somatic mutations or changes in DNA methylation (Fig. 1).

Approaches to detect somatic mutations in cfDNA initially gained traction with tumour-informed assays for monitoring of patients during therapy. On the basis of initial studies of individual mutations in cfDNA^{31–33}, these approaches involved sequencing at deep coverage ($\geq 30,000\times$) of a specific gene or a few genes of interest and identifying tumour-specific (somatic) mutations in cfDNA of patients who were known to have these alterations in their corresponding tumours^{34–36} (Fig. 1). In the context of early cancer detection, the mutations present

Table 1 | Cancer types with recommended screening

Cancer type	Screening guidelines or programmes	Target population	Screening test(s)
Universal age-based screening			
Breast ¹⁵⁰	Multiple national or international guidelines (USPSTF and others)	Generally starting at age 40–50 years and ending at age 69–74 years	Mammography
Colorectum ¹⁵⁰	Multiple national or international guidelines (USPSTF and others)	Generally starting at age 45–55 years and ending at age 74 years	Colonoscopy, flexible sigmoidoscopy, FIT, FIT-DNA, gFOBT, CT colonography
Cervix ¹⁵⁰	Multiple national or international guidelines (USPSTF and others)	Generally starting at age 18–29 years and ending at age 60–70 years	Cytology, HPV
Universal screening in high-risk groups*			
Lung ¹⁵¹	Multiple national or international guidelines (USPSTF and others)	Generally starting at age 50–55 years and ending at age 70–80 years; ≥20–30 pack-year smoking history, with some requiring current smoking or having quit smoking within the last 10–15 years	LDCT
Liver ¹⁵²	Multiple national or international guidelines (AASLD and others)	Individuals with cirrhosis or HBV or HCV infection and other high-risk characteristics	Ultrasound, AFP, DCP, AFP-L3
Geographic-based screening (with example screening programmes)			
Stomach ¹⁵³	Screening programmes in several Asian countries	Generally starting at age 40 years	Endoscopy, serum-pepsinogen test, barium tests
Head and neck (nasopharyngeal carcinoma) ¹⁵⁴	Cancer screening programme in rural areas in China	Individuals at age 30–69 years with high-risk characteristics	EBV serology test followed by nasopharyngoscopy
Oesophagus ¹⁵⁴	Screening programmes in China	Individuals at age 45–74 years with high-risk characteristics	Endoscopy
Mixed recommendations on screening			
Skin ¹⁵⁰	Some guidelines (for example, Federal Joint Committee in Germany) recommending screening whereas others have concluded evidence is insufficient (for example, USPSTF)	No general consensus	Visual skin examination by a physician, self-examination
Prostate ¹⁵⁰	Some guidelines recommend screening whereas others recommend against screening	Starting at age 40–50 years if recommended	PSA, DRE

Cancer types are ordered within each category by prevalence¹. AASLD, American Association for the Study of Liver Diseases; AFP, α-fetoprotein; AFP-L3, *Lens culinaris* agglutinin-reactive fraction of AFP; CT, computed tomography; DCBE, double-contrast barium enema; DCP, des-γ-carboxyprothrombin; DRE, digital rectal examination; EBV, Epstein–Barr virus; FIT, faecal immunochemical test; gFOBT, guaiac faecal occult blood test; HBV or HCV, hepatitis B or C virus; HPV, human papillomavirus; LDCT, low-dose computed tomography; PSA, prostate-specific antigen; USPSTF, US Preventative Services Task Force. *Individuals with specific high-risk characteristics such as certain genetic variants may have different screening recommendations. For example, endometrial cancer screening by means of endometrial biopsy is recommended by the American Cancer Society for individuals 35 years or older that have or are at risk for hereditary non-polyposis colorectal cancer¹⁵⁵. Another example is screening for pancreatic cancer, which has been recommended for high-risk individuals with >5% lifetime risk, including those individuals with a family history of cancer or who carry specific mutations¹⁵⁶.

in the tumour of an individual are not known, and therefore a panel of genes commonly mutated in cancers may be used for targeted sequencing^{37–40} (Table 2). As an example, the targeted error correction sequencing approach described the capture of 58 cancer-related genes followed by deep sequencing at approximately 30,000× coverage and error correction through redundant sequencing and bioinformatics filters³⁷.

Extensive efforts have been made to reduce error rates from targeted sequencing, typically relying on barcoding strategies coupled with consensus sequence formation using multiple reads of the same molecule, as well as bioinformatics filters, effectively reducing error rates from library preparation and sequencing from 10^{-3} to $<10^{-7}$ (ref. 41). However, other sources of non-tumour-specific (background) sequence changes can confound these analyses, including somatic mutations in white blood cells that may be clonally expanded, resulting in clonal haematopoiesis of indeterminate potential (CHIP) which may be detected in cfDNA^{42–44}. Machine learning approaches such as Lung-CLIP that incorporate features of mutated fragments that differ

between ctDNA and non-cancer-derived cfDNA in targeted regions may enrich for tumour-derived mutations³⁹.

A universal challenge facing targeted sequencing approaches has been the low number of diploid genome equivalents per millilitre of blood (~1,500), which limits the sensitivity of detection for any specific alteration, whether mutation or methylation in origin⁴⁵. Converging lines of analysis have led to the realization that broad genome-wide assessment of genomic or epigenomic alterations would provide increased sensitivity even using limited amounts of cfDNA^{46,47}. In contrast to targeted sequencing and analyses of alterations to cfDNA fragments, genome-wide approaches could enable investigation of broader compendium of ctDNA alterations across the genome (Fig. 1). Improved potential performance, decreasing costs of WGS, and simpler laboratory processes provide advantages of this method compared with those involving targeted enrichment.

Detection of genome-wide structural variants. Genome-wide analyses of cfDNA for cancer detection were initially performed over a

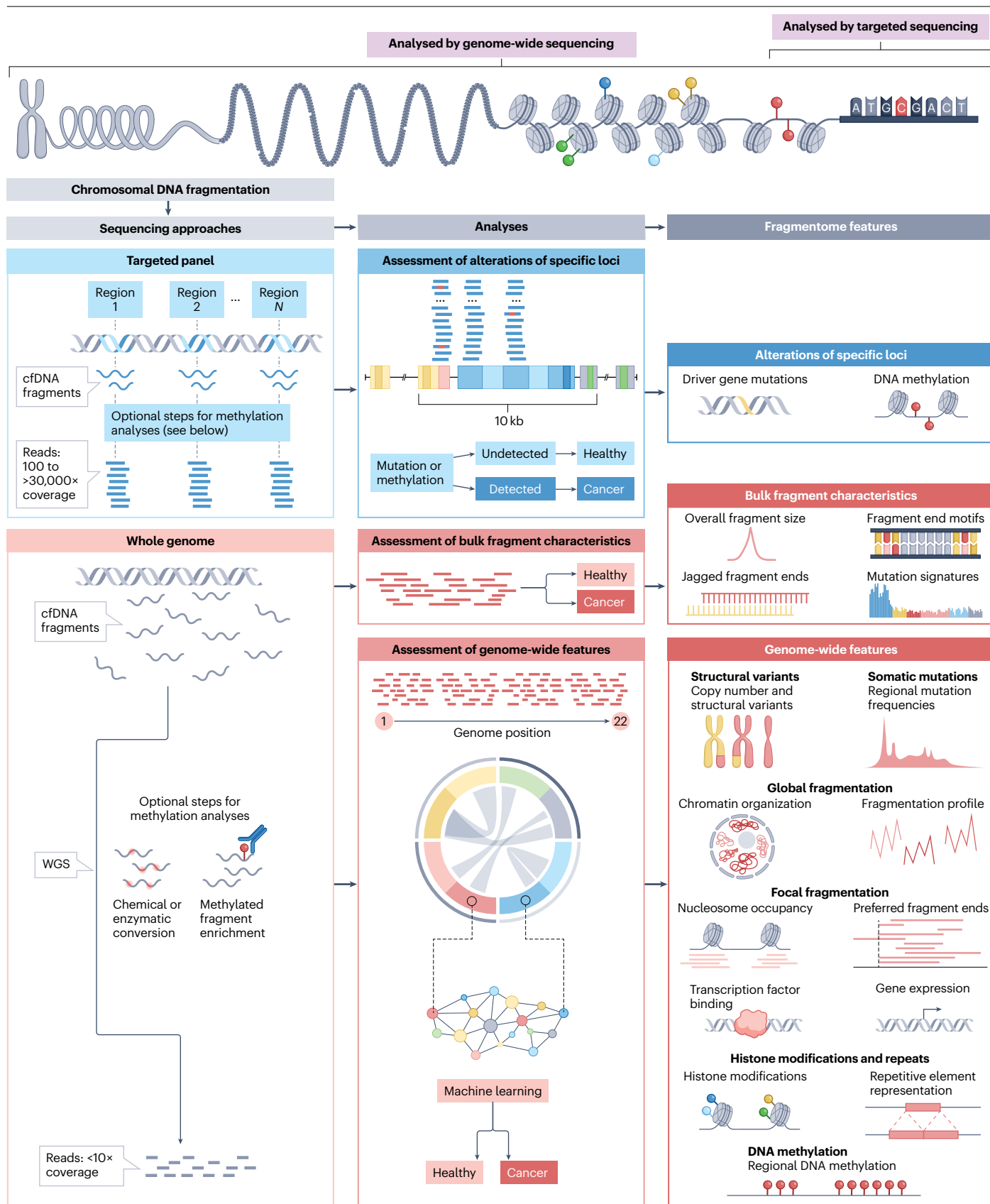


Fig. 1 | cfDNA features and analysis for early detection of cancer.

Chromosomal DNA fragmentation is modulated by chromatin organization, including large-scale open and closed chromatin and smaller-scale nucleosome positioning, as well as other genomic and epigenomic characteristics, giving rise to the cfDNA fragmentome. Approaches for analysis of the cfDNA fragmentome have included targeted sequencing and genome-wide sequencing. Targeted sequencing involves enriching cfDNA fragments for specific sites in the genome and deep sequencing (for example, 100 to >30,000× sequence coverage) for the detection of mutations in cancer driver genes or changes in DNA methylation. Whole-genome sequencing (WGS) at low coverage (<10×, with optional steps for methylation detection) and analysis of the entire genome has enabled detection of an increasing compendium of alterations. Genome-wide features are typically analysed using machine learning models to develop classifiers that predict whether an individual has cancer. Early detection approaches that utilize these features are listed in Tables 2 and 3. When selecting a method for cfDNA analysis, each approach offers distinct advantages and disadvantages. Targeted

approaches may improve the signal-to-noise ratio by focusing on cancer-specific alterations and using established error correction strategies. However, they require additional steps for target enrichment and deep sequencing, with performance limited by cfDNA input quantity and the need for matched white blood cell sequencing for mutation-based analyses. Whole-genome bulk analyses do not require target enrichment and are not limited by cfDNA input, but performance can be limited owing to small cancer-specific differences, few assessed features, and no positional information. Genome-wide analyses measure numerous features across the genome, combining positional insights with machine learning for high performance. They avoid target enrichment, require minimal cfDNA input, and enable biomarker identification using existing data sets. Disadvantages include the need for complex statistical approaches including machine learning to integrate many features and reliance on less-established error-suppression methods. For some methylation analyses, additional steps for bisulfite or enzymatic treatment of cfDNA are required with either approach.

decade ago for identification of tumour-specific structural changes⁴⁸. Using a high-throughput approach for quantifying sequence representation across the genome, called digital karyotyping⁴⁹, or through paired analyses of rearranged ends (PARE) to identify tumour-specific rearrangements⁵⁰, these studies have revealed that chromosomal gains or losses, as well as rearrangements, were detectable in cfDNA from individuals with cancer without prior knowledge of the alterations in the tumour⁴⁸ (Fig. 2). Additional studies showed that copy number changes were broadly detectable genome-wide in cfDNA from patients with a variety of cancers^{51–53}. As cells contain multiple copies of the mitochondrial genome and the abundance of mitochondrial DNA (mtDNA) can be altered in cancers⁵⁴, changes in mtDNA copy number has been reported in cfDNA from patients with several cancer types^{47,55,56}. New methods to detect chromosomal abnormalities emerged, including plasma aneuploidy score⁴⁸ or machine learning-based detection of chromosomal representation using DNA evaluation of fragments for early interception (DELFI)⁴⁷. Additional approaches utilizing copy number alterations for quantifying levels of ctDNA, such as ichorCNA³⁷ or DELFI tumour fraction⁵⁸, can be used for noninvasive monitoring of tumour load in patients with cancer.

Genome-wide analyses of somatic mutations. Somatic mutation frequencies in cancer genomes can vary by orders of magnitude between cancer types and individual tumours, but they usually involve thousands to tens of thousands of single-nucleotide variants, hundreds to thousands of insertions or deletions (indels), and tens to hundreds of larger structural rearrangements (>100 bp in size) per tumour²⁰. Declining sequencing costs coupled with methodological advances have enabled genome-wide analyses of the compendium of somatic mutations in the circulation even at single-molecule (1×) sequence coverage. One challenge that is shared with targeted approaches has been filtering of non-tumour-specific ‘background’ sequence changes occurring during library preparation, sequencing or alignment, somatic mutations from non-cancer cells, or germline variants^{59–61}. Single-molecule mutation frequencies may also be influenced by variable error rates across samples and sequencing instruments, confounding comparisons of mutation frequencies between patients^{60,62,63}.

A single-molecule WGS method using patient-specific regional differences in mutation frequencies (that is, controlled within each individual) reduced sequencing-associated variability in mutation frequency estimates and improved detection of alterations in ctDNA⁶⁰. This approach, called GEMINI, could be used to identify altered regional

mutation profiles in cfDNA from individuals with cancer (Table 2 and Fig. 2). The regional mutational profiles across the genome reflected differences in replication timing, gene expression, histone modifications, and chromatin organization and could be used to detect cancer and identify the tissue of origin⁶⁰. These studies were conducted using standard library preparation and low-coverage (<5×) WGS and may, therefore, enable facile integration with approaches utilizing other genome-wide features such as cfDNA fragmentation^{60,61}.

Genome-wide analyses of repetitive elements. Repetitive elements, including long interspersed nuclear elements, short interspersed nuclear elements, long terminal repeats, transposable elements, and human satellite families, make up more than half of the human genome⁶⁴ and have been broadly implicated in cancers^{65–67}. These sequences have historically been difficult to study as their repetitive nature often precludes unique mapping of short sequence tags to their genomic origin, limiting analyses in cfDNA to specific repeat regions with common polymorphisms⁶⁸ or microsatellites flanked by sufficiently unique sequence^{69,70}. Long-read sequencing improves mapping to repetitive regions of the genome⁷¹; however, because most cfDNA fragments are small (<200 bp), mapping of these sequences does not benefit from long-read sequencing instruments.

To overcome these challenges, an alignment-free approach called ARTEMIS has been developed⁶⁷, identifying ~1.2 billion *k*-mers (sequences of length *k*) that can uniquely identify 1,280 repeat element types using typical WGS. Through analyses of WGS data from tumour tissues and cfDNA, the approach identified hundreds of altered repeat elements that had not been previously implicated across cancer types that were associated with structural variation and epigenetic states (Fig. 2). Machine learning models incorporating repeat element landscapes using direct analyses of cfDNA have detected individuals with early-stage lung and liver cancers, and may be used to improve tissue-of-origin determination in a multi-cancer screening setting⁶⁷.

Epigenetic characteristics of cfDNA

Targeted DNA methylation-based approaches. Following early studies demonstrating that epigenetic changes were detectable in cfDNA⁷², methods involving targeted methylation analyses of cfDNA have been developed for early cancer detection^{73–77} (Fig. 1). These methods have historically detected methylated sites after treatment with bisulfite, a chemical compound that converts cytosine but not 5-methylcytosine (5mC) to uracil, which after PCR amplification gets

Table 2 | Representative cfDNA-based early cancer detection approaches

Overall approach	Method or study name	Assay	Features analysed	Initial cancer types analysed
Targeted approaches	TEC-Seq ³⁷	Targeted hybrid-capture NGS of 58 genes comprising 80.93 kb	Somatic mutations	Breast, colorectum, lung, ovary
	CancerSEEK ³⁸	PCR-based targeted sequencing of a 61-amplicon panel, immunoassay for eight proteins	Somatic mutations, proteins	Breast, colorectum, oesophagus, liver, lung, ovary, pancreas, stomach
	Lung-CLiP ³⁹	Targeted hybrid-capture NGS of 266 genes comprising 355 kb	Somatic mutations, CNVs	Lung
	RealSeqS ¹⁵⁷ , A-PLUS ¹⁵⁸	PCR-based targeted sequencing of ~350,000 repetitive elements	Chromosomal CNVs, Alu element representation	12 cancer types
	Galleri ^{75,a}	Targeted hybrid-capture bisulfite sequencing of 17.2Mb	DNA methylation	>50 cancer types
	PanSEER ⁷⁴	Bisulfite conversion followed by targeted PCR to amplify and sequence 595 genomic loci encompassing 11,787 CpG sites		Colorectum, oesophagus, liver, lung, stomach
	ELSA-seq ⁷⁶	Bisulfite conversion and hybrid-capture of cfDNA covering 80,672 CpG sites comprising approximately 1.05Mb		Lung
	Shield ^{133,a}	Barcoding of methylated and unmethylated cfDNA fragments followed by targeted hybrid-capture of ~1Mb	Somatic mutations in APC or KRAS, DNA methylation, fragment end positions	Colorectum
	SPOT-MAS ⁷⁷	Targeted hybrid-capture bisulfite sequencing of 450 regions encompassing 18,000 CpG sites and WGBS	DNA methylation, fragment end motifs, genome-wide fragmentation profile and CNVs	Breast, colorectum, stomach, lung, liver
	Helzer et al. ¹⁵⁹	Targeted hybrid-capture sequencing of 2.1–2.4Mb	Fragment size at first coding exons	Bladder, breast, lung, prostate
Bulk analyses	Mouliere et al. ⁹³	Low-coverage WGS	Fragment size, CNVs	Bile duct, brain, breast, colorectum, kidney, ovary, pancreas, skin
	Motif diversity score ⁹⁴		Fragment end motifs	Colorectum, head and neck, liver, lung
	Pointy ⁵⁹		Somatic mutation signatures	Breast, colorectum, lung, ovary, pancreas, stomach
	Chan et al. ⁵²	WGBS	Bulk methylation	Breast, liver, lung, nasopharynx, smooth muscle
	Jaggedness index ⁹⁵		Fragment jaggedness	Liver
Genome-wide approaches	Genome-wide structural variants			
	Plasma aneuploidy score from digital karyotyping ⁴⁸	Low-coverage WGS	Chromosomal CNVs	Breast, colorectum
	PARE ⁴⁸		Rearrangements	Breast, colorectum
	plasma-Seq ⁵³		CNVs	Prostate
	Chan et al. ⁵²	WGBS	CNVs	Breast, liver, lung, nasopharynx, smooth muscle
	Genome-wide global fragmentation			
	DELFI ^{47,115,118,127}	Low-coverage WGS	Fragmentation profile, chromosomal CNVs, mtDNA representation	Breast, bile duct, colorectum, liver, lung, ovary, pancreas, stomach
	FirstLook Lung ^{128,a}			Lung
	Wang et al. ¹³⁰ , Zhang et al. ¹²⁹		Fragmentation profile, CNVs, and fragment end motifs from bulk cfDNA analyses	Liver, lung
	Genome-wide focal fragmentation			
	Jiang et al. ¹²¹	Low-coverage WGS	Fragment end positions	Liver
	E-index ¹²³			Bile duct, brain, breast, colorectum, liver, lung, ovary, pancreas, stomach

Table 2 (continued) | Representative cfDNA-based early cancer detection approaches

Overall approach	Method or study name	Assay	Features analysed	Initial cancer types analysed
Genome-wide approaches (continued)	Ulz et al. ¹¹³		TFBS coverage	Colorectum
	Griffin ¹¹⁶			Breast, bile duct, colorectum, lung, ovary, pancreas, stomach
	LIQUORICE ¹¹⁴		Coverage at DNase I-hypersensitive sites	Ewing sarcoma
	GALYFRE ¹²²		Fragment end positions, fragment end motifs from bulk cfDNA analyses	10 cancer types
	CRAG ¹⁴⁰		Coverage and size at fragmentation hot spots	Bile duct, breast, colorectum, liver, lung, ovary, pancreas, stomach
	Bae et al. ⁶¹		Coverage and size at nucleosome-depleted regions, somatic mutation profiles	10 cancer types
	Stanley et al. ¹⁴¹		Nucleosome spacing in the first 10kb of genes	Breast, colorectum, blood
Genome-wide somatic mutations				
	GEMINI ⁶⁰	Low-coverage WGS	Somatic mutation profiles	Liver, lung
Genome-wide DNA methylation				
	cfMeDIP-seq ⁴⁶	MeDIP optimized for cfDNA followed by sequencing	DNA methylation	Breast, bladder, blood, colorectum, kidney, lung, pancreas
	HCC methylation score ⁹⁰	Single-molecule real-time sequencing		Liver
	cfTAPS ⁸⁴	TAPS optimized for cfDNA	DNA methylation, fragment size	Liver, pancreas
	FRAGMA ¹²⁵	Low-coverage WGS	DNA methylation signals from fragment end motifs	Liver
	Noë et al. ¹⁰⁷		DNA methylation signals from fragment end positions	Pancreas
	AlphaLiquid Screening ⁸⁷	EM-seq	DNA methylation, fragmentation profile, CNVs	Colorectum, liver, lung, prostate
	EMMA ¹³¹	WGBS		Oesophagus
Genome-wide repeats				
	ARTEMIS ⁶⁷	Low-coverage WGS	Repetitive elements, coverage at regions enriched for certain histone marks or epigenetic states	Liver, lung

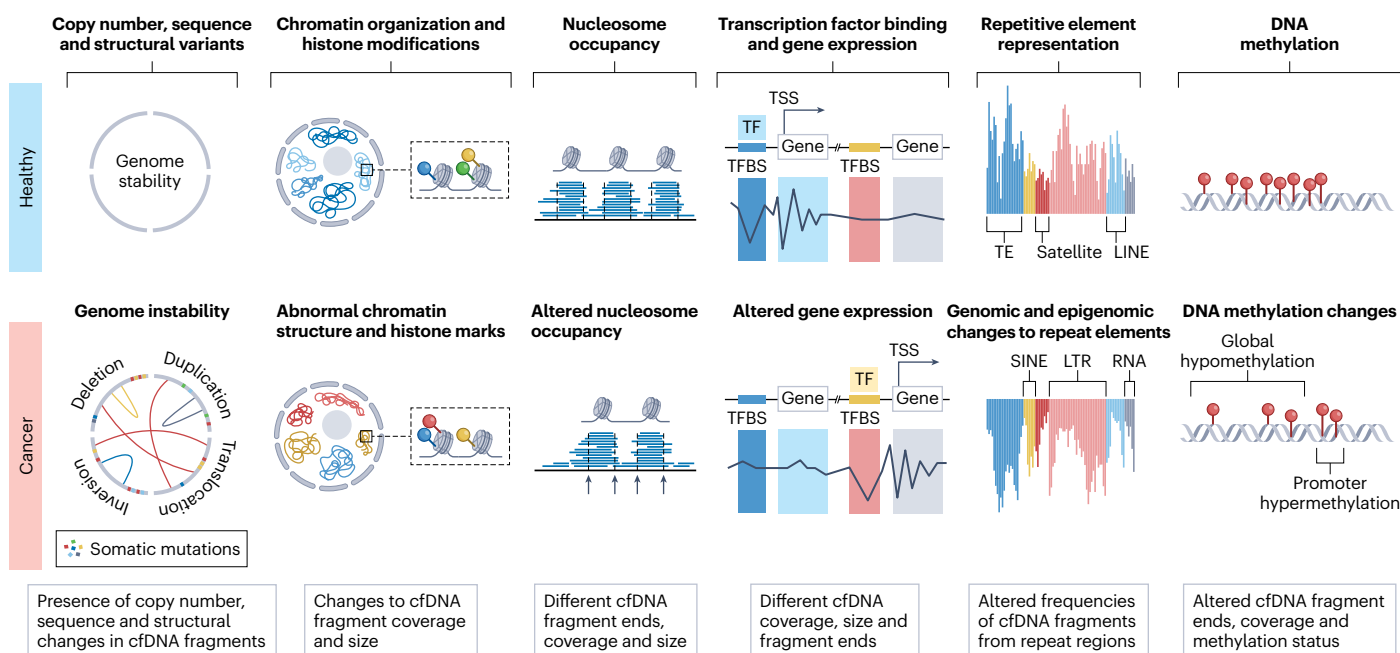
cfDNA, cell-free DNA; CNV, copy number variation; EM-seq, enzymatic methyl sequencing; MeDIP, methylated DNA immunoprecipitation; NGS, next-generation sequencing; PARE, paired analyses of rearranged ends; TAPS, TET-assisted pyridine borane sequencing; TEC-Seq, targeted error correction sequencing; TFBS, transcription factor binding site; WGBS, whole-genome bisulfite sequencing; WGS, whole-genome sequencing. ^aLiquid biopsy screening tests that have been clinically validated.

converted to thymine in newly synthesized DNA strands⁷⁸. A clinically used cfDNA targeted methylation assay is the Galleri test that uses hybridization capture of 17.2 Mb of genome and sequencing of these regions at moderate coverage (139×), enabling detection of different solid or haematological cancer types⁷⁵. The sensitivity of this approach across cancer types by stage was 18% for stage I, 43% for stage II, 81% for stage III, and 93% for stage IV at high specificity (>99%)⁷⁵ (Table 3). In a real-world study, the method detected only 8% of patients with cancer at stage I across all solid tumours and 9% of patients with cancer across all stages of lung cancer⁷⁹. Thus, overall sensitivity and, in particular, detection of stage I disease remain a challenge with this approach. Multiple factors can contribute to the low detection rates for early-stage disease with this assay and related methods. First, ctDNA

concentrations in many patients with cancer are often low⁸⁰, a universal challenge of cfDNA-based early detection that is further hampered in this case by the targeted nature of the assay (<1% of the genome) in which ctDNA fragments outside these regions will be missed. Additionally, ctDNA fragments may be degraded or removed during bisulfite treatment or sample preparation. Whether the challenges faced by this approach for cancer detection are specific to the methodology or will be similar for other targeted methylation-based detection approaches remains to be seen.

Genome-wide analyses of cfDNA methylation. One challenge with bisulfite treatment of DNA followed by NGS has been degradation of a substantial portion of DNA⁸¹. To overcome this issue, bisulfite-free

a Altered cfDNA characteristics in patients with cancer



b Examples of genome-wide alterations in cfDNA fragmentomes in patients with cancer

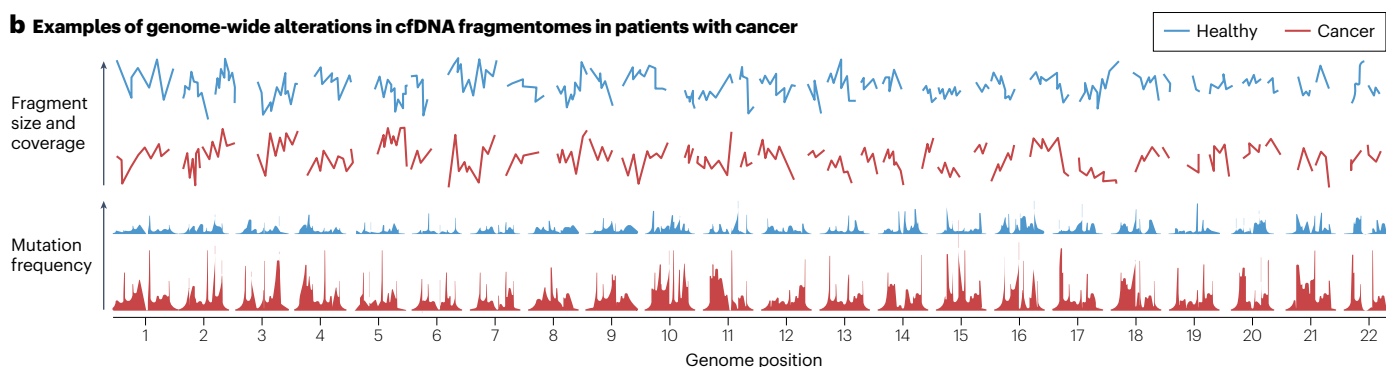


Fig. 2 | Differences in genome-wide cfDNA characteristics between individuals with and without cancer. a, Compared to normal cell-free DNA (cfDNA), circulating tumour DNA (ctDNA) from cancer cells may contain tumour-derived structural variants, including deletions, duplications, inversions and translocations, as well as cancer-associated somatic mutations. Large-scale alterations to chromatin organization in cancer genomes associate with altered fragment coverage and sizes across the genome, with fewer and smaller fragments typically associated with open, active chromatin and more and larger fragments with closed, inactive chromatin. Cell-type or cancer-associated differences in nucleosome positioning lead to regional differences in fragment end positions, coverage and size in cfDNA. Fragments may also be enriched or

depleted and have altered size and end positions at specific transcription factor binding sites (TFBS) and transcription start sites (TSS) depending on the genes that are active in the cancer cells giving rise to ctDNA. Structural and epigenetic changes may also alter the frequency of repetitive elements in the circulation. ctDNA is generally globally hypomethylated but may also have hypermethylation of specific sites such as promoters. **b,** Genome-wide cfDNA characteristics are analysed with respect to their position in the genome, exemplified by regional fragment size and regional mutation frequencies. LINE, long interspersed nuclear element; LTR, long terminal repeat; SINE, short interspersed nuclear element; TE, transposable element.

methods for the detection of methylation in cfDNA have been developed. A derivative of the methylated DNA immunoprecipitation (MeDIP) approach⁸² optimized for cfDNA, called cfMeDIP-seq, involves immunoprecipitation of cell-free methylated DNA across the whole genome followed by sequencing⁴⁶. This approach was applied to 189 plasma samples from 7 cancer types and to a validation cohort of 199 samples, demonstrating high performance for the detection of multiple solid or haematological cancer types⁴⁶. A benefit of this approach

was the simultaneous analysis of multiple methylated regions, providing a theoretically higher sensitivity than targeted approaches involving a limited number of methylated regions. A follow-up analysis of 608 blood samples suggested potentially high performance of the approach for the detection of brain cancers and determination of tumour subtypes⁸³. A methylation-based approach, cfTAPS, uses TET-assisted pyridine borane sequencing (TAPS) treatment before PCR, which causes both 5mC and 5-hydroxymethylcytosine (5hmC) to be

Table 3 | Clinically validated cfDNA tests for cancer detection in the USA

Sponsor	Assay	Cancer type	Trial identifier	Study name	Study type	Patient characteristics	Enrolment (n)	Sensitivity by cancer stage ^a (%)					Specificity ^a (%)	NPV ^a (%)
								APL	I	II	III	IV		
Delfi Diagnostics Inc.	FirstLook	Lung cancer	NCT04825834 (ref. 160)	L101	Prospective case-control	High risk for lung cancer	958	NA	71	89	88	98	58.0	99.8
								NA	16	27	36	75	29	98.6
Guardant Health, Inc.	Shield	Colorectal cancer	NCT04136002 (ref. 162)	ECLIPSE	Prospective cohort	Eligible for colorectal cancer screening	22,877	13	65	100	100	100	90.0	99.9
								NA	6,662	Asymptomatic, age >50 years	100	100	83	99.9

Tests with published clinical validation data within a clinical trial that are available to patients in the USA. APL, advanced precancerous lesion; NA, not available; NPV, negative predictive value. ^aObserved or stage-adjusted performance obtained from indicated studies.

amplified as a thymine base^{84,85}. Using this approach, altered DNA methylation was detected in regulatory regions in cfDNA from patients with hepatocellular carcinoma and pancreatic ductal adenocarcinoma, and the method could distinguish individuals with cancer from those without⁸⁴. Another approach, EM-seq, used a series of enzymatic reactions to convert unmodified cytosines to uracil, which is replaced with thymine during PCR, enabling identification of sites with 5mC or 5hmC from as little as 100 pg of DNA⁸⁶. Using this technique to analyse 950 plasma samples, a platform incorporating whole-genome methylation with other features indicated high sensitivity and specificity for the detection of colon cancer (*n* = 107), liver cancer (*n* = 113), lung cancer (*n* = 238) and prostate cancer (*n* = 131)⁸⁷.

An emerging area is direct detection of DNA methylation in cfDNA using an altered electrical signal in nanopore sequencing^{88,89} and single-molecule real-time sequencing using altered polymerase kinetics surrounding CpG sites⁹⁰. Analyses of sequencing data from these long-read technologies has been found to broadly recapitulate features of cfDNA derived from short-read sequencing data, including copy number variation (CNV) profiles, nucleosome profiles, fragment end motifs, and inferred ctDNA concentrations^{91,92}. Whereas short-read technologies are typically limited to analyses of fragments <600 bp⁹⁰, long-read sequencing with these approaches enables analysis of longer cfDNA fragments (for example, >10 kb) that may contain a larger set of informative CpG sites that could be helpful for assessing the tissue of origin of a given fragment⁹⁰. Although few studies have been conducted to date, initial efforts show promise for cancer detection using methylation derived from long-read sequencing⁹⁰.

Analyses of cfDNA fragmentation

Bulk cfDNA measurements. Alongside advances in detecting tumour-derived variants in cfDNA, there have been extensive efforts to develop methods for cancer detection based on general characteristics of cfDNA fragments, largely using overall summaries or ‘bulk’ cfDNA measurements that are irrespective of genomic position. Examples include approaches which measure the overall size distribution of cfDNA⁹³, the frequency of specific fragment end sequence motifs⁹⁴, or the jaggedness of cfDNA fragments⁹⁵, as well as overall methylation levels^{52,96} or mutation signatures⁵⁹ (Fig. 1).

Approaches to measure the overall size distribution of cfDNA benefited from fundamental work on the properties of cell-free DNA fragment sizes in patients with cancer or healthy controls observed through gel electrophoresis^{97,98}, electron microscopy^{99,100} and PCR^{101–103}. These previous studies have subsequently enabled, using NGS-based methods, a refined characterization of the overall size distribution of cfDNA fragments^{55,104,105}. In individuals with cancer, there is a slight shift towards shorter sizes of cfDNA fragments^{26,103,104,106}, and enriching for shorter cfDNA fragments has been shown to improve detection of ctDNA⁹³. Reasons for the observed overall decrease in cfDNA sizes in patients with cancer can include global hypomethylation and increased gene expression¹⁰⁷, which are common features of cancer genomes¹⁰⁸.

Other bulk properties of cfDNA have emerged using NGS-based methods that may be useful for cancer detection. Fragment end sequence motifs appear to be altered in individuals with cancer⁹⁴. cfDNA from patients with hepatocellular carcinoma and several other cancer types showed a higher diversity of the 256 possible 4-mer fragment end motifs compared to individuals without cancer as quantified by a motif diversity score⁹⁴. Another potentially useful property of most cfDNA are single-stranded ends termed ‘jagged ends’, as increased jaggedness has been observed in individuals with hepatocellular carcinoma⁹⁵.

Both cfDNA jaggedness and end motif composition can be modulated by cleavage by specific nucleases^{95,109}, but the role of such nucleases in human cancer remains unclear. For all such bulk analyses, caution must be used given the sensitivity of cfDNA fragments to rapid degradation and the potential that small changes in bulk populations, including in fragment ends, may result from differences in pre-analytical conditions of samples from different patient populations¹¹⁰.

Well-established characteristics of tumour genomes have led to additional methods for cancer detection using bulk cfDNA analyses. For example, using low-coverage WGS, the frequency of specific genomic sequence alterations that reflect mutagenic processes during tumour evolution, termed ‘mutational signatures’, has been directly identified in cfDNA from individuals with cancer⁵⁹. Bulk epigenomic analyses of DNA methylation using whole-genome bisulfite sequencing have revealed global hypomethylation of cancer genomes that was detectable in plasma from individuals across a wide range of cancer types, including hepatocellular carcinoma, breast cancer, lung cancer, nasopharyngeal cancer, smooth muscle sarcoma, and neuroendocrine cancer as well as in urine of patients with bladder cancer^{52,96}.

Although recent methods for bulk measurements of cfDNA using NGS may benefit from analysis of many fragments, such approaches ignore additional information from genome-wide position-dependent differences that may exist between individuals with and without cancer. For example, in individuals with cancer, cfDNA mutation frequencies were higher in heterochromatic regions of the genome where DNA repair may be impaired⁶⁰. Overall, although bulk cfDNA approaches typically reflect small changes and none of these are currently used in clinically approved tests, they have helped pave the way towards understanding features of cfDNA.

Focal analyses of cfDNA fragmentation at regulatory sites. It is now emerging that cfDNA fragmentation reflects chromatin structures and epigenetic changes across the genome. For example, it was demonstrated that cfDNA fragment coverage reflects nucleosome positions in the tissue of origin, providing a link between nucleosome arrangement and the pattern of cfDNA fragmentation¹¹¹. Furthermore, fragmentation profiles (a combination of coverage and size of cfDNA) have been found to closely reflect chromatin compartmentalization⁴⁷. In addition, the sites of modified chromatin structure and transcription factor binding identified in epigenomic studies of cancer tissues have been shown to be associated with altered cfDNA fragmentation^{61,112–118}. As a result, high-resolution analyses of cfDNA fragments have enabled inference of expressed genes^{112,119}, transcription factor binding^{113,115,118}, nucleosome-depleted regions¹²⁰, and large-scale chromatin organization⁴⁷ in the tissue of origin (Fig. 2). Transcription factor binding may also directly protect short cfDNA fragments (35–80 bp) from degradation as shown using single-stranded library preparation and NGS¹¹¹, providing yet another avenue for evaluating cfDNA fragments which may be altered in cancer.

Focal analyses of cfDNA involve measuring the quantity or characteristics of cfDNA fragments at small sites (for example, <1 kb) at specific locations that together typically comprise less than a few percent of the genome. Examples of focal features that have been evaluated for early detection of cancer include cfDNA coverage at transcription factor binding sites (TFBS) or other regulatory regions^{113–116,118}, or the frequency of fragment ends at specific genomic locations^{107,121–123} (Table 2). Methods involving focal analyses of cfDNA fragments differ from targeted sequencing in that they do not involve target enrichment before sequencing. Instead, information from WGS has typically been

used to assess cfDNA features at specific areas in the genome. Using low-coverage WGS (<10× coverage), a single focal feature may be covered by only a few cfDNA reads at any single region, and therefore these analyses have involved aggregating data from multiple regions into a meta-locus to derive more stable measurements.

An analysis of aggregated coverage at binding sites for 504 transcription factors enabled the identification of changes at TFBS and detection of a subset of individuals with early-stage colorectal cancers¹¹³. In an analysis of individuals with small-cell lung cancer (SCLC) in which the ASCL1 pioneer transcription factor is highly expressed, there was a prominent decrease in coverage aggregated across ~13,000 ASCL1-binding sites, enabling the detection of SCLC and distinguishing this subtype from non-small-cell lung cancer¹¹⁵. Similarly, in a cohort of individuals with liver cancer, altered coverage at transcription factor binding sites linked to liver cancer facilitated its detection in a screening population¹¹⁸. Although most research has focused on detecting cancers in adults, analyses of cfDNA from paediatric sarcoma patients using low-coverage WGS have revealed reduced fragment coverage at Ewing sarcoma-related DNase I-hypersensitive sites, transcription factor binding sites, and enhancers, allowing machine learning-based detection of patients with this condition¹¹⁴. Collectively, these studies highlight the potential to measure cancer-associated focal transcriptome alterations using cfDNA fragment abundances directly from cfDNA, which may be enhanced by adjusting for GC content-related coverage biases inherent to NGS experiments^{116,124}.

Studies have also revealed distinct sites across the genome that are enriched or depleted for fragment ends in cfDNA from patients with cancer. These were initially found to be altered in liver cancers¹²¹ but have since been identified in many cancer types and have been associated with changes in DNA methylation and gene expression^{107,122,123}. The proportion of fragment ends at methylated CpG sites has been shown to be increased compared to unmethylated ones^{107,125}. Additionally, higher levels and larger sizes of cfDNA fragments have been independently associated with regions of CpG methylation^{107,123,126} and reduced gene expression¹⁰⁷. Incorporating the distribution of cfDNA fragment end positions at CG and CCG sites in cfDNA-based cancer detection models enabled distinguishing individuals with pancreatic cancer from healthy individuals¹⁰⁷.

Methods that utilize focal cfDNA features benefit from the fact that these features are often preferentially altered in specific cancer types. Therefore, they can provide insights into the tissue of origin or cancer subtype, which has implications for treatment and prognosis. However, as a result, features that are important for the detection of one cancer type may be suboptimal for the detection of other cancer types, necessitating multiple features and models for different clinical scenarios. Although focal approaches may have potentially strong signals at epigenetic hotspots, they often only utilize a small subset of the total sequencing data (for example, <1%) and may benefit from combination with genome-wide approaches that incorporate additional information¹¹⁸.

Genome-wide analyses of global cfDNA fragmentation. Global cfDNA fragmentation approaches involve measuring the quantity or characteristics of cfDNA fragments across the entire genome, without restricting analyses to specific sites of interest. Although analyses of targeted, bulk or focal changes in cfDNA provided a limited view of overall genomic alterations, it became evident over time that cfDNA fragment characteristics vary across the genome and can be simultaneously assessed using a genome-wide approach⁴⁷. This approach was

particularly appealing as examining these characteristics across the genome increased the information content from low-coverage WGS, enabling the theoretical detection of many more alterations. As a result, it increased the sensitivity of these methods, wherein focused analyses of limited sites would be insufficient. One example is the DELFI method, which involves shallow WGS (1–2× coverage) of cfDNA⁴⁷. Fragments are analysed with respect to their size and distribution in regions or ‘bins’ across the genome, creating ‘fragmentation profiles’ that can be compared between individuals (Fig. 2). In healthy individuals, cfDNA fragmentation profiles correlated with nucleosome distances, DNA fragmentation patterns of nuclease-treated nuclei from healthy lymphocytes, and chromatin conformation capture (Hi-C) data for open and closed compartments of lymphoblastoid cells⁴⁷. In patients with cancer, fragmentation profiles appeared to result from a combination of chromatin changes and chromosomal and other large-scale genomic alterations in the neoplastic cells. A machine learning algorithm was trained to recognize the differences in these profiles and other features between individuals with and without cancer, enabling prediction of whether a new individual has cancer. In this initial description of the approach, DELFI was shown to detect seven types of cancer, including cancers of the bile duct, breast, colorectum, stomach, lung, ovary and pancreas, suggesting its potential to identify multiple types of cancer

and their tissue of origin. Follow-up studies focusing on detection of specific cancers in patient populations at increased risk have demonstrated high performance in detecting lung cancer¹¹⁵, liver cancer¹¹⁸ or ovarian cancer¹²⁷. These proof-of-concept studies paved the way for the larger prospective studies in intended use populations, which clinically validated the cfDNA fragmentome FirstLook blood test for enhancing lung cancer screening¹²⁸. This study has demonstrated that the fragmentome-based blood test had the needed performance characteristics, including a negative predictive value (NPV) of 99.8%, to be clinically useful¹²⁸ (Table 3). Variations of the genome-wide fragmentation approach have been incorporated in a number of other classifiers in development for noninvasive cancer detection^{77,87,114,129–131}.

Early cancer detection in at-risk populations

In principle, cfDNA-based tests could be used to improve existing screening programmes or to detect cancers for which no current screening modalities exist. Tests for early detection of cancer should ideally be accessible, affordable and scalable to large populations of individuals (Box 1). The considerations for implementation of single-cancer early-detection tests vary compared to those for multi-cancer early-detection tests and may also differ for individuals with cancer predisposition syndromes. Here, we briefly discuss the

Box 1 | Important characteristics of cancer screening tests

Proof-of-concept studies, also known as discovery, are an initial step in the pathway to develop cell-free DNA (cfDNA) biomarkers and have important limitations. Studies used in discovery are often based on convenience samples that may have pre-analytical factors that are not fully known, or they may include participants that do not fall within the intended use population of the test. A subsequent case–control study with prospectively collected samples is an opportunity to refine the classifier with uniform collection and processing of samples in the intended use population. Classifiers developed from these case–control studies have a higher chance of succeeding in large clinical and analytical validation studies that are necessary for laboratory-developed tests.

Sensitivity

How often a test can identify early-stage cancers is crucial for any screening approach. For both single-cancer and multi-cancer early-detection tests that evaluate cancers for which screening is known to have a survival benefit, a high sensitivity (true-positive rate) and low false-negative rate is imperative. Otherwise, such tests risk falsely reassuring individuals that they are cancer-free and could discourage participation in established screening programmes.

Specificity

Maintaining a high specificity (true-negative rate) and a low false-positive rate is especially important for balancing the benefit-to-harm ratio in multi-cancer screening tests aimed at average-risk populations, in which the prevalence of cancer is low and the risk of complex and unnecessary follow-up diagnostic evaluations is high. Single-cancer tests in at-risk populations may tolerate a moderate specificity as follow-up tests typically involve already recommended screening approaches.

Positive predictive value

Of those with a positive cancer screening test, the proportion that have cancer (positive predictive value (PPV)) should be high to reduce harm from screening. High PPVs (for example, >10%) are important in screening for cancers with low prevalence if the follow-on diagnostic procedure is invasive and can be harmful to patient health. Lower PPVs (for example, <5%) may be acceptable if those testing positive are encouraged to receive already recommended follow-on noninvasive screening tests such as imaging.

Negative predictive value

Of the individuals with a negative cancer screening test, the negative predictive value (NPV) is the proportion that are cancer-free. In cancers for which there is a demonstrated clinical benefit for regular screening, a high NPV provides confidence that an individual with a negative test is cancer-free.

Scalability

Translating a high-performance approach to widespread clinical use poses important technical challenges. Simple laboratory procedures involving limited and robust reagents, equipment and laboratory processes are easier to scale.

Accessibility

Screening tests involving a standard blood draw taken at a routine physical exam may increase uptake compared to approaches necessitating travel to specialized sites that use imaging or require an invasive procedure. The cost of these tests needs to be low for feasibility on a population scale to avoid exacerbating existing health disparities.

practicalities related to implementation of cfDNA-based early detection and highlight the challenges and opportunities of these clinically validated tests.

Focused early cancer detection in at-risk populations

Early cancer detection tests have typically been used to identify a specific cancer type in a population at increased risk, such as individuals of a certain age (for example, 50–80 years of age) and/or with certain exposures (for example, tobacco smoke or hepatitis B virus) (Table 1). Single-cancer blood tests can be optimized for identifying such cancers in which there is already sufficient evidence that early detection provides a net benefit. For example, it has been suggested that a liquid biopsy could serve as a screening method for augmenting LDCT imaging for lung cancer detection in high-risk individuals^{60,115,128}. In this setting, the liquid biopsy approach would need to have high sensitivity to not deter individuals from seeking LDCT owing to a false-negative result. However, a moderate specificity may be acceptable as a positive test would result in the patient being encouraged to obtain follow-on screening they were already recommended to receive (that is, LDCT). The combined specificity and positive predictive value of a positive liquid biopsy pre-screen followed by a positive LDCT result would be higher than for LDCT alone¹¹⁵, potentially reducing the number of unnecessary invasive diagnostic procedures. In settings where resources are limited, a liquid biopsy screen for asymptomatic or even symptomatic individuals could enrich the population for individuals who tend to benefit from further investigation^{128,132}. As an example, an individual with a positive test result using the FirstLook blood test for lung cancer would receive LDCT imaging of the lungs, whereas a negative test would suggest follow-up annual screening¹²⁸.

Clinical studies evaluating cfDNA-based tests for other cancer types have also been conducted (Table 3). The recently developed Shield test for the detection of colorectal cancer identified 11 out of 17 (65%) individuals with stage I disease, and all individuals with stage II disease ($n = 14$), stage III disease ($n = 17$), or stage IV disease ($n = 10$) at a false-positive rate of 10.1%¹³³. The detection rate of the assay among 1,116 participants with advanced precancerous lesions was 13.2% (95% confidence interval: 11.3–15.3), slightly above the false-positive rate of the assay. The low sensitivity for the detection of stage I disease and of precancerous lesions compared to stool testing or colonoscopy highlights the challenges of this approach for colorectal cancer screening. As another example of a clinically validated cfDNA test, detection of Epstein–Barr virus DNA by quantitative PCR for the identification of nasopharyngeal carcinoma has resulted in a high performance for the detection of this disease¹³⁴, highlighting that viral biomarkers, although not applicable for most cancer types, may be effective in specific settings.

Cancer predisposition syndromes

Germline variants in specific genes, such as *BRCA1* or *BRCA2*, confer an exceptionally high risk of developing one or several types of cancer, often at an unusually early age, requiring intense prevention and surveillance measures for affected individuals¹³⁵. cfDNA-based early detection approaches may be useful for cancer detection in individuals with cancer predisposition syndromes. In a longitudinal cohort of 89 individuals with Li–Fraumeni syndrome, an integrated cfDNA analysis of somatic mutations, CNVs, fragmentation, and DNA methylation from targeted sequencing, shallow WGS or cfMeDIP-seq has identified cancer-associated signals in select individuals earlier than conventional screening¹³⁶. Another example is altered DNA fragmentation

as measured by low-coverage WGS in patients with PTEN hamartoma tumour syndrome¹³⁷. Given the high probability of individuals with cancer predisposition syndromes developing cancer, a cfDNA-based test on its own would need to have high sensitivity in this population to avoid false reassurance of being cancer-free owing to a false-negative test (Box 1). It is conceivable that cfDNA-based single-cancer or multi-cancer tests could be integrated in the future into existing surveillance programmes. Additional studies will need to assess whether incorporating these tests would yield improved patient outcomes, considering the treatment options currently available.

Multi-cancer early detection

In principle, multi-cancer cfDNA detection tests could be used for screening. However, an important consideration for multi-cancer tests is which cancer types to test for, as with current treatment options only a subset of cancers have convincingly demonstrated benefits from early detection. Widespread screening for cancer types that have no or weakly established net benefits from screening has the potential for substantial harm owing to overdiagnosis and/or over-treatment. A glaring example of overdiagnosis comes from an analysis of prostate-specific antigen testing for prostate cancer in the USA, using data from 1986 to 2005. This study suggested that over a million men were overdiagnosed and received unnecessary treatment, such as with surgery and radiation, for prostate cancer¹³⁸. Beyond the anxiety, financial strain and other challenges associated with a cancer diagnosis, prostate cancer treatments can lead to long-term side effects including incontinence, impotence and other complications¹³⁹.

When screening for multiple cancer types in an average-risk, asymptomatic population with relatively low cancer prevalence, it will be crucial to maintain a high specificity to minimize the risk of false-positive results and their negative consequences. Sensitivity to detect early precancerous lesions and early-stage disease must also be high to avoid providing false reassurance, which could discourage individuals from participating in established screening programmes for common cancers. Current clinically available multi-cancer approaches for early-stage lung, colon and breast cancers do not have sensitivities or NPVs high enough to detect or rule out disease, respectively, to be considered useful for screening in these common cancer types⁷⁹.

An important component of any positive multi-cancer early detection test is pinpointing the probable tissue of origin of a cancer signal, thereby reducing the potential for a diagnostic odyssey. Epigenetic alterations captured through DNA methylation and fragmentation changes are particularly attractive for localizing cancers to their tissue of origin as these types of alterations may vary markedly between tissues^{46,47,61,75,84,87,115,140,141}. Results from a clinical study involving the methylation-based Galleri assay indicated that although overall sensitivity for cancer detection was low, of the 35 out of 121 (29%) cancers that were detected, 29 (83%) were assigned to the correct anatomic location, indicating a relatively high predictive accuracy⁷⁹. Other cfDNA approaches that analyse genetic alterations such as mutations in driver genes, mutation signatures or regional mutation frequencies, and/or structural variants that all differ by cancer type to varying extents, can also be useful for tissue-of-origin analyses^{38,47,59,60}.

As earlier detection of cancers does not always translate to improved outcomes, exemplified, for instance, by increased ovarian cancer diagnosis through screening that did not lead to a survival benefit¹⁴², clinical trials evaluating survival benefit from blood-based cancer screening are crucial. Higher performing tests, as well as insights derived from comprehensive molecular tumour characterization

Box 2 | Other applications of cfDNA-based assays

In addition to early detection of cancer, there are several other clinical scenarios in which cell-free DNA (cfDNA)-based assays are being evaluated. Here, we highlight three important areas, namely therapeutic stratification, disease monitoring, and detection of minimal residual disease.

Therapeutic stratification

Decades of studies in cancer genomics have identified somatic mutations in tumours that are associated with response or resistance to therapy, and these alterations are now being detected from liquid biopsies¹⁶³. As an example, liquid biopsy approaches involving deep targeted sequencing of specific regions of the genome have been designed to detect mutations, copy number changes, and fusions such as those implicated in sensitivity or resistance to targeted therapies involving the *EGFR*, *KRAS*, *BRAF*, *ALK*, *BRCA2*, *PIK3CA* or *ESR1* genes among others, as well as microsatellite instability^{23,69,70,164}. One challenge is that when ctDNA concentrations are low, liquid biopsies may miss alterations known to be present in the tumour tissue. Conversely, liquid biopsies may detect additional subclonal alterations that were missed by a single tumour biopsy^{37,163}.

Disease monitoring

For late-stage patients with cancer, the increased burden of circulating tumour DNA (ctDNA) in plasma can be an early indication of resistance to treatment that often precedes clinical detection through imaging^{165,166}. The ability to switch treatments at the first

molecular signs of progression for improving overall survival is being evaluated in clinical trials¹⁶⁷. Owing to clonal haematopoiesis of indeterminate potential (CHIP), parallel sequencing of white blood cells has been recommended by several studies^{43,44}. The detection of altered genome-wide fragmentation patterns appears promising because the approach does not rely on a small number of mutations that may be subclonal, and CHIP mutations do not appear to affect characteristics of the fragmentome^{44,47}. Larger studies using both targeted and genome-wide approaches are needed for a more definitive comparison of these approaches.

Detection of minimal residual disease

The fundamental challenge faced by blood-based approaches for the detection of remaining cancer after potentially curative treatment, termed minimal residual disease, is that ctDNA may comprise less than 0.01% of cfDNA molecules circulating in plasma¹⁶⁸. In this setting, the most sensitive approaches have been ultra-deep sequencing using targeted mutation-based approaches in combination with sequencing of the primary tumour tissue to guide detection of clonal mutations in the tumour¹⁶⁸. Tumour-informed genome-wide approaches may overcome challenges related to the low number of genome equivalents in plasma obtained from a standard blood draw and provide potentially more sensitive methods^{50,169}. In the future, plasma-only genome-wide approaches may enable de novo detection of tumour-derived DNA without requiring availability of tumour tissue.

and improvements in cancer treatment, could increase associations between stage-shifting and overall survival.

Future perspectives

cfDNA-based approaches for early cancer detection and for other applications (Box 2) have dramatically advanced in recent years and yet continue to evolve, with new technologies opening up possibilities for improved performance. Newer sequencing technologies are enabling higher-throughput analyses of cfDNA, including interrogation of relatively understudied long cfDNA fragments with long-read sequencing technologies¹⁴³. PCR-free library preparation, using high-fidelity polymerases, or single-molecule redundant sequencing strategies¹⁴⁴ could potentially improve performance of methods analysing genome-wide sequence changes. New approaches are enabling simultaneous analysis of both sequence and methylation changes in the same DNA fragments^{145,146}. Studies have suggested that combining complementary analyses, such as protein biomarkers or targeted sequence mutations with cfDNA fragmentation, may improve performance^{47,127}. Recently developed strategies to pharmacologically increase cfDNA abundance may improve the performance of targeted sequencing-based approaches¹⁴⁷. Improved risk stratification using polygenic risk scores¹⁴⁸ may improve test performance by enriching the population for those with higher risk for cancer¹⁴⁹. Additional efforts and trials are needed to continue to improve these approaches and validate these tests for clinical use.

For newly developed approaches, accurate determination of test performance will need to await independent clinical validation. Comparing performance for unvalidated tests is typically problematic as studies

outside of prospective trials are subject to confounding owing to variability of pre-analytical factors, assessment of symptomatic individuals, potential batch effects, and the possibility of overfitting through multiple evaluations of the analysed data. Improved understanding of biological changes in ctDNA characteristics from different cancer types in asymptomatic populations, combined with analytical and clinical validation of newly developed tests, would shed light on the approaches that have the highest tendency to succeed. Assessments of costs and scalability will ultimately be crucial to determine which methods can be widely accessible for screening programmes nationally and globally.

A critical question is whether ctDNA-based screening approaches can increase overall survival for patients with cancer types beyond those where screening is recommended. Achieving this would require ctDNA testing to detect cancer earlier or in individuals not currently screened using conventional approaches, and this early detection would need to translate into more effective therapies. Although clinical trials aimed to demonstrate an overall survival benefit may take years to complete, their results could ultimately determine the clinical value of these technologies and their role in cancer management.

Conclusions

Decades of research uncovering the existence and origins of cfDNA, as well as the numerous unique properties of tumour-derived ctDNA, coupled with substantial technological advances to study nucleic acids, have resulted in an explosion of blood-based assays that promise to aid early cancer detection. Although initial liquid biopsy approaches focused on deep targeted sequencing of a small subset of gene regions, the advances in technology now enable comprehensive analyses that

Glossary

Apoptosis

A form of programmed cell death that can be initiated by either extracellular or intracellular mediators in response to a predefined developmental programme or in response to cellular stress.

Cancer screening

Population-scale testing of asymptomatic individuals for cancer.

Cell-free DNA

(cfDNA). DNA fragments that are not contained within cells, which can be produced through apoptosis, necrosis and active secretion.

cfDNA fragmentome

The genome-wide compendium of cfDNA fragments in the circulation, providing an integrated view of the chromatin, genome, epigenome and transcriptome states of normal and cancer cells of an individual.

Chromatin

A complex of DNA and proteins that can exist as open, 'active' chromatin termed euchromatin, or as heterochromatin, which refers to closed 'inactive' chromatin.

Circulating tumour DNA

(ctDNA). cfDNA fragments from tumour cells released into the bloodstream.

Classifiers

Algorithms that assign an entity to one or more groups.

Clonal haematopoiesis of indeterminate potential

(CHIP). A genetically distinct population of blood cells caused by somatic mutations in blood cell progenitors.

Diagnostic odyssey

When the process to diagnose a disease is long and difficult, often requiring multiple tests and procedures.

False-negative rate

The probability that an individual with cancer will test negative, which is equivalent to 1-sensitivity.

False-positive rate

The probability that an individual without cancer will test positive, which is equivalent to 1-specificity.

Germline variants

Genetic variants that were already present in the germline leading to its presence in all cells throughout the body.

Li–Fraumeni syndrome

A rare disorder resulting from pathogenic germline variants in the *TP53* gene that greatly increases the risk of developing cancer.

Lymphoblastoid cells

Immortalized B lymphocytes generated by infection with Epstein–Barr virus.

Massively parallel sequencing

Often referred to as next-generation sequencing (NGS), this laboratory approach is used to determine the sequence of a large number of DNA fragments simultaneously.

Microsatellites

Short stretches of DNA wherein one to several nucleotides are repeated multiple times.

Necrosis

A form of cell death distinct from apoptosis; necrosis can be induced by irreversible cell injury and is characterized by breakdown in the plasma membrane and release of intracellular contents.

Negative predictive value

(NPV). The probability that an individual with a negative screening test does not have cancer.

Nucleosomes

Segments of DNA wrapped around histone proteins; a mononucleosome refers to a single nucleosome, whereas a di-nucleosome or tri-nucleosome refers to two or three nucleosomes.

Pack-year

A quantification of the smoking history of an individual determined by multiplying the number of packs of cigarettes smoked per day by the number of years of smoking.

Polygenic risk scores

(PRS). Scores derived from multiple genetic variants related to the risk of an outcome.

Polymorphisms

Variations in DNA sequence at specific positions in the genome that contribute to genetic variation and might alter the structure, function or expression of the gene product.

Positive predictive value

(PPV). The probability that an individual with a positive screening test has cancer.

Pre-analytical factors

Factors that occur before sample analysis in the laboratory that may influence test results, such as how samples are collected and stored, which have been shown to influence ctDNA levels.

PTEN hamartoma tumour syndrome

A rare genetic condition caused by germline mutations in the *PTEN* gene with multiple health implications including an increased risk of developing certain types of cancer.

Repetitive elements

Sequences of DNA that are repeated multiple times in the genome.

Sensitivity

The probability that an individual with cancer will test positive, which can be estimated directly from a case–control study as the proportion of cancers that test positive out of all cancers in the study (also known as the true-positive rate).

Specificity

The probability that an individual without cancer will test negative, which can be estimated directly from a case–control study as the proportion of individuals without cancer that test negative out of all individuals without cancer in the study (also known as the true-negative rate).

Stage-shifting

Detecting cancers at an earlier stage of the disease owing to screening.

Structural variants

Large genomic rearrangements that are typically ≥ 50 bp.

capture a growing compendium of cancer-related alterations across the genome. These advances have resulted in widespread interest in liquid biopsies for noninvasive genomic profiling of cancers, disease monitoring in late-stage patients, detection of minimal residual disease, and now early detection of cancer.

However, several important challenges remain. The vast majority of studies described here have analysed individuals with a confirmed

cancer diagnosis or those whose blood samples were collected close to the time of diagnosis. In the context of screening asymptomatic individuals with potentially less advanced cancers, performance of cfDNA tests may be lower. The sensitivity of a screening test will need to be high even in early-stage disease so that individuals without a positive result will not be deterred from utilizing other existing screening approaches. Additionally, although the focus on performance is

important, the cost-efficiency and accessibility of these tests will be crucial for widespread adoption. Balancing the creative urge to develop increasingly complex laboratory tests with the necessity of sensitive, scalable and accessible approaches will enable individuals worldwide to benefit from these efforts.

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Author contributions

All authors researched data for the article, contributed substantially to the discussion of the content, wrote the article, and reviewed and/or edited the manuscript before submission.

Competing interests

D.C.B., Z.H.F., J.P. and R.B.S. are inventors on patent applications submitted by Johns Hopkins University related to cell-free DNA analyses. J.P. and R.B.S. are founders of Delfi Diagnostics, and R.B.S. is a consultant for this organization. V.E.V. is a founder of Delfi Diagnostics, serves on the board of directors, and owns Delfi Diagnostics stock, which is subject to certain restrictions under university policy. Additionally, Johns Hopkins University owns equity in Delfi Diagnostics. V.E.V. divested his equity in Personal Genome Diagnostics (PGDx) to LabCorp in February 2022. V.E.V. is an inventor on patent applications submitted by Johns Hopkins

University related to cancer genomic and cell-free DNA analyses that have been licensed to one or more entities, including Delfi Diagnostics, LabCorp, Qiagen, Sysmex, Agios, Genzyme, Esoterix, Ventana and ManaT Bio. Under the terms of these license agreements, the University and inventors are entitled to fees and royalty distributions. V.E.V. is an adviser to Viron Therapeutics and Epitepe. These arrangements have been reviewed and approved by the Johns Hopkins University in accordance with its conflict-of-interest policies.

Additional information

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